

BuildManager: End-to-End Workflow & Navigation Journey

This document provides a comprehensive step-by-step workflow for the **BuildManager** Construction Project Monitoring System. It describes the complete user journey, starting from the initial entry point, detailing the module hierarchy, available actions, and features on each screen.

1. Initial Entry Point: Authentication

The user journey begins at the login screen, ensuring secure access to the tool based on predefined roles.

Login & Registration Page (/login)

- **Purpose:** Secure entry gateway.
 - **Features & Actions:**
 - **Sign In:** Users enter their registered **Email** and **Password** to access the system.
 - **Password Visibility Toggle:** An eye icon to show/hide the password.
 - **Create Account (Registration):** New users can toggle to the registration form to provide their **Full Name**, **Email**, **Password**, and request a specific **Role** (Admin, Manager, Engineer, Viewer).
 - **Redirection:** Upon successful authentication, the user is redirected to the **Dashboard (/)**.
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2. Core Navigation (Sidebar)

Once authenticated, the user is presented with the main application layout. The primary navigation tool is the **Sidebar**, which dynamically adjusts based on the user's role (Admin, Manager, Engineer, Viewer).

The Sidebar is divided into logically grouped functional areas:

1. **Overview**
 2. **Projects**
 3. **Resources (3M)**
 4. **Usage Tracking**
 5. **Finance**
 6. **Billing & Expenses**
 7. **System**
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3. Section Breakdown & Screen Flow

3.1 Overview Section

This section provides a high-level summary of the construction business and immediate notifications.

- **Dashboard (/)**
 - **Audience:** All Users.
 - **Features:** Displays key performance indicators (KPIs), active project counts, recent alerts, and summary charts. Acts as the command center.
- **Alerts (/alerts)**
 - **Audience:** All Users.
 - **Features:** A dedicated feed for system notifications, budget warnings, or missing usage logs.
- **Recycle Bin (/recycle-bin)**
 - **Audience:** Admins & Managers.

- **Features:** A safe-delete mechanism. Users can view soft-deleted records (e.g., deleted projects or materials) and choose to **Restore** them or **Permanently Delete** them.

3.2 Projects Section

This is the core operational area for managing construction sites and teams.

- **Projects Catalog** (`/projects`)

- **Features:** Lists all active and past projects.

- **Actions:** Users can *Create, Edit, Delete, or Search* for projects. Clicking on a specific project redirects to the **Project Detail** page.

- **Project Detail** (`/projects/:id`)

- **Features:** A deep-dive screen for a single project. Contains sub-tabs or sections summarizing specific resources, financials, and team members assigned exclusively to this project.

- **Progress** (`/project-progress`)

- **Features:** Gantt charts or milestone tracking detailing completion percentages of different project phases.

- **Team** (`/project-team`)

- **Audience:** Admins & Managers.

- **Features:** Assigns registered users to specific projects, managing who has operational oversight over which site.

- **Import Project** (`/import`)

- **Audience:** Admins & Managers.

- **Features:** A bulk-upload module. Users can upload `.xlsx` (Excel) files to ingest large volumes of historical data. The module previews errors and prevents duplicate data entries before saving.

3.3 Resources (3M) Section

Manages the foundational catalogs of the three M's of construction: Materials, Machines, and Manpower.

- **Materials** (`/materials`)

- **Features:** A master catalog of all physical materials (e.g., Cement, Steel).
- **Actions:** Add new items, update unit types (kg, tons), and set baseline costs.

- **Machines** (`/machines`)

- **Features:** Equipment inventory (e.g., Excavators, Cranes).
- **Actions:** Track equipment ownership, rental status, and operational capacity.

- **Workers** (`/workers`)

- **Features:** A database of laborers and site staff.
- **Actions:** Define worker trades/skills and daily wage rates.

3.4 Usage Tracking Section

The daily operational logbook for engineers and site managers.

- **Material Usage** (`/material-usage`)

- **Features:** Logs daily consumption of materials on specific sites.
- **Actions:** Select a project, select a material, enter quantity consumed, and log the date.

- **Manpower Usage** (`/manpower-usage`)

- **Features:** Daily attendance and labor allocation.
- **Actions:** Log how many workers (or specific teams) were present at a site and calculate daily labor costs.

- **Machine Usage** (`/machine-usage`)

- **Features:** Equipment utilization tracking.
- **Actions:** Record the hours logged by specific machinery on a project to calculate fuel consumption and wear-and-tear costs.

3.5 Finance Section

Restricted to higher-level management to track capital inflow and debt.

- **Investors (/investors) & Financiers (/financiers)**
 - **Features:** CRM-style profiles for entities providing capital or loans.
 - **Actions:** Add contact details and track relationship history.
- **Investments (/investments)**
 - **Features:** Records of equity capital infused into the business or specific projects.
- **Loans (/loans)**
 - **Features:** Tracking borrowed capital, interest rates, and principal amounts.
- **Interest Payments (/interest-payments)**
 - **Features:** A ledger for scheduling and recording interest payouts against active loans.

3.6 Billing & Expenses Section

Manages the outflow of cash and client invoicing.

- **Expenses (/expenses)**
 - **Features:** Day-to-day operational expenditures (e.g., fuel, petty cash, permit fees).
 - **Actions:** Log expense amounts, categorize them, and attach them to specific projects.
- **Billing (/billing)**
 - **Features:** Client invoice generation and tracking.
 - **Actions:** Create bills based on project milestones, track payment statuses (Pending, Paid).
- **Budget Analysis (/budget-comparison)**
 - **Features:** A reporting dashboard.

- **Actions:** Visually compares the estimated budget of a project against actual expenses and resource usage to identify cost overruns.

3.7 System Section

Strictly restricted to Administrators for platform maintenance.

- **Users** (`/users`)

- **Features:** Complete user management.

- **Actions:** Approve new registrations, change user roles, reset passwords, or deactivate accounts.

- **Audit Log** (`/audit-log`)

- **Features:** A security and compliance trail.

- **Actions:** View a chronological list of all system actions (e.g., "User X deleted Project Y"), including timestamps and IP tracking.

4. Example Step-by-Step User Journey

Scenario: A Project Manager setting up a new construction site.

1. **Authentication:** The manager logs in via `/login`.
2. **Dashboard Review:** Lands on the Dashboard (`/`) to review current active projects and alerts.
3. **Project Creation:** Navigates to **Projects** (`/projects`), clicks "Add New Project", and fills out the site details.
4. **Team Assignment:** Navigates to **Team** (`/project-team`) and assigns a Site Engineer to the newly created project.
5. **Resource Verification:** Checks the **Materials** (`/materials`) catalog to ensure the necessary resources (e.g., specific grade concrete) are available in the system.
6. **Usage Monitoring:** Over the next week, the assigned Engineer logs into the system and uses the **Usage Tracking** section (`/material-usage` , `/manpower-usage`) to

log daily consumption.

7. **Financial Review:** The Manager navigates to **Budget Analysis** (</budget-comparison>) to review if the Engineer's daily usage logs are keeping the project within the estimated financial limits.